

MelodyMemo: An Emotion-Aware Memory–Music Modeling and Recommendation System for Older Adults

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PURPOSE & MOTIVATION

61M

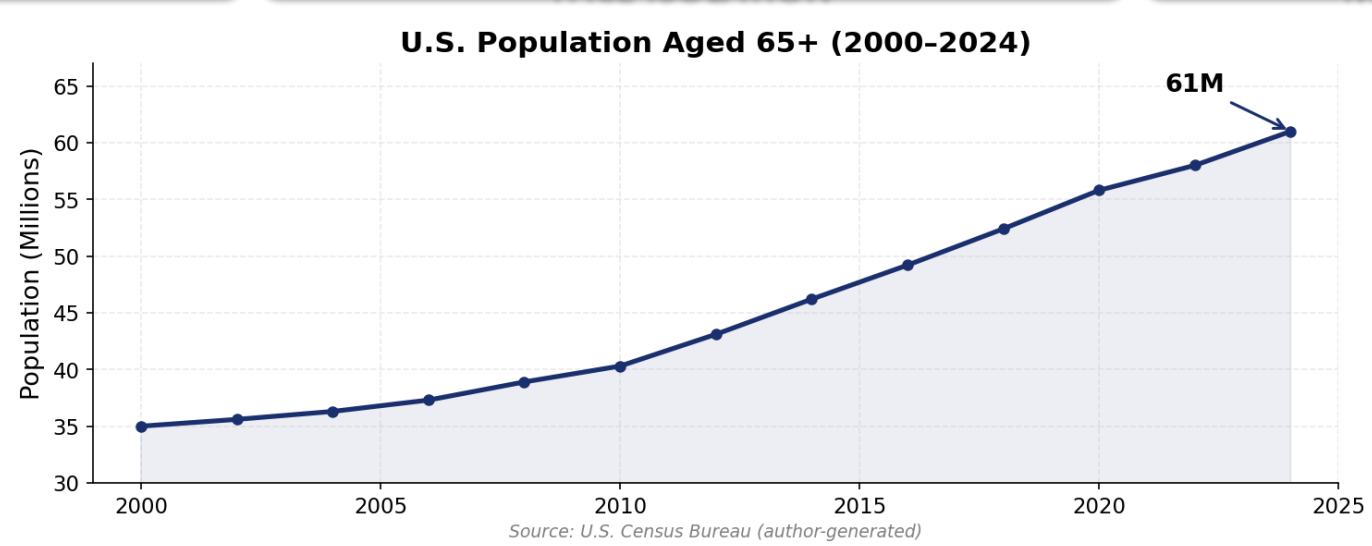
AMERICANS 65+

>25%

FACE ISOLATION



MUSIC UNLOCKS MEMORY



- 61M Americans are 65+; fastest-growing demographic in the U.S. (Census Bureau, 2023).
- Social **isolation** affects >25%, linked to **cognitive decline** and **depression**
- Music-Evoked Autobiographical Memories (**MEAMs**) remain robust even in early-stage cognitive impairment
- No prior system combines narrative analysis with emotionally matched music

That's the gap MelodyMemo aims to fill.

RESEARCH QUESTIONS & HYPOTHESES

RQ1: Can a pre-trained transformer model reliably extract emotions from autobiographical narrative text?

RQ2: Can a hybrid recommendation algorithm outperform individual algorithms in preserving emotional alignment?

H1: DistilRoBERTa achieves $\geq 75\%$ emotion classification accuracy across seven categories.

H2: Hybrid (collaborative + content-based) outperforms each individual algorithm and a random baseline.

METHODOLOGY

R1: Emotion Extraction

- Compiled 50 autobiographical stories (100–300 words each)
- Each story labeled with one of 7 emotions
- Classified using pre-trained DistilRoBERTa transformer
- Cross-validated against GPT-4o (90%) and Gemini 2.5 Flash (88%)
- Extracted entities with spaCy NER

R2: Music Recommendation

- 50 songs (1939–1975) with tempo, valence, energy, genre
- 25 senior user profiles (ages 76–88), ~80% sparse
- 4 algorithms: random, collaborative (Jaccard), content-based (cosine), hybrid (50/50)

50

STORIES

50

SONGS

25

USERS

~80%

SPARSITY

EMOTION → MUSICAL FEATURE MAPPING

JOY

EXAMPLE



Fast

TEMPO

0.90

VALENCE

High

ENERGY

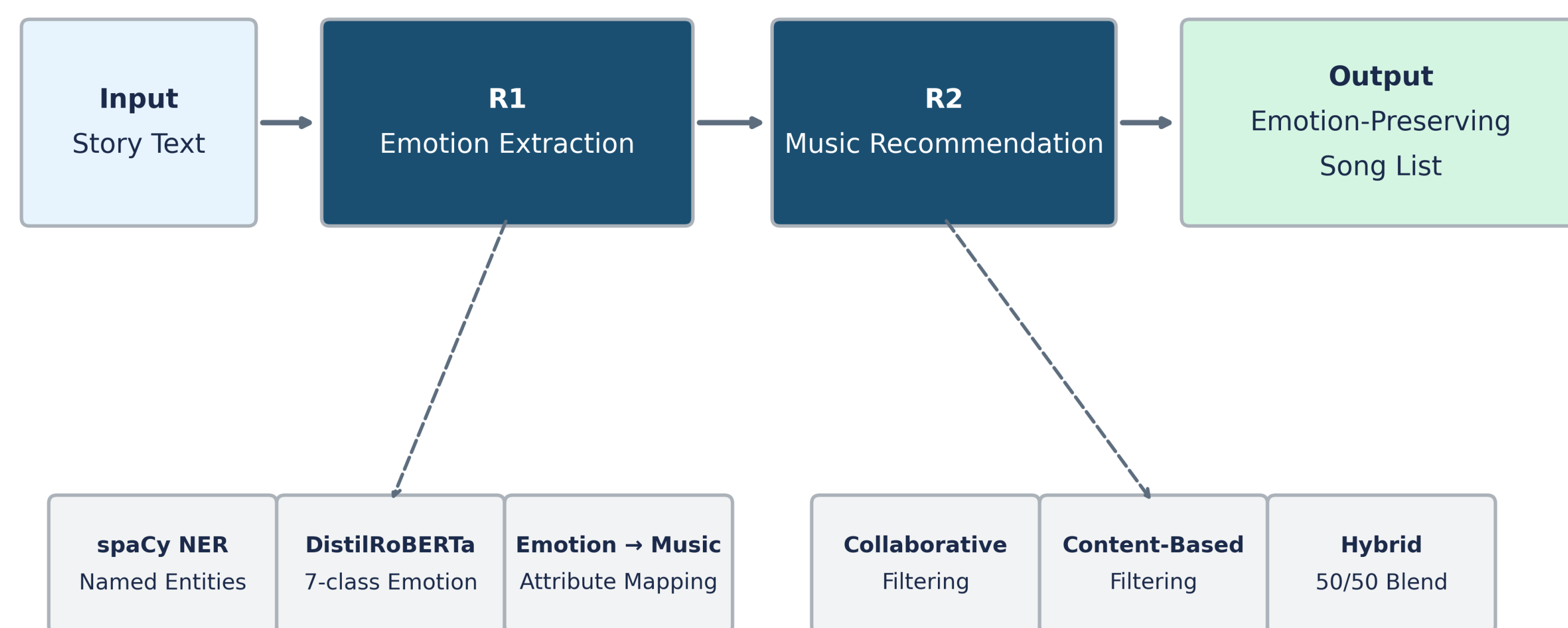
Major

MODE

Emotion	Tempo	Valence	Energy	Mode
Joy	130 BPM	0.90	0.90	Major
Sadness	70 BPM	0.25	0.30	Minor
Anger	130 BPM	0.30	0.90	Minor
Fear	100 BPM	0.30	0.60	Minor
Love	100 BPM	0.85	0.60	Major
Surprise	130 BPM	0.70	0.80	Major
Neutral	100 BPM	0.50	0.60	Major

HOW IT WORKS

MelodyMemo System Architecture



R1 Module Components

R2 Algorithm Variants

Story Input → Emotion Detection (R1) → Music Mapping → Hybrid Recommendation (R2) → Playlist

7

EMOTIONS

50

STORIES

50

SONGS

25

USERS

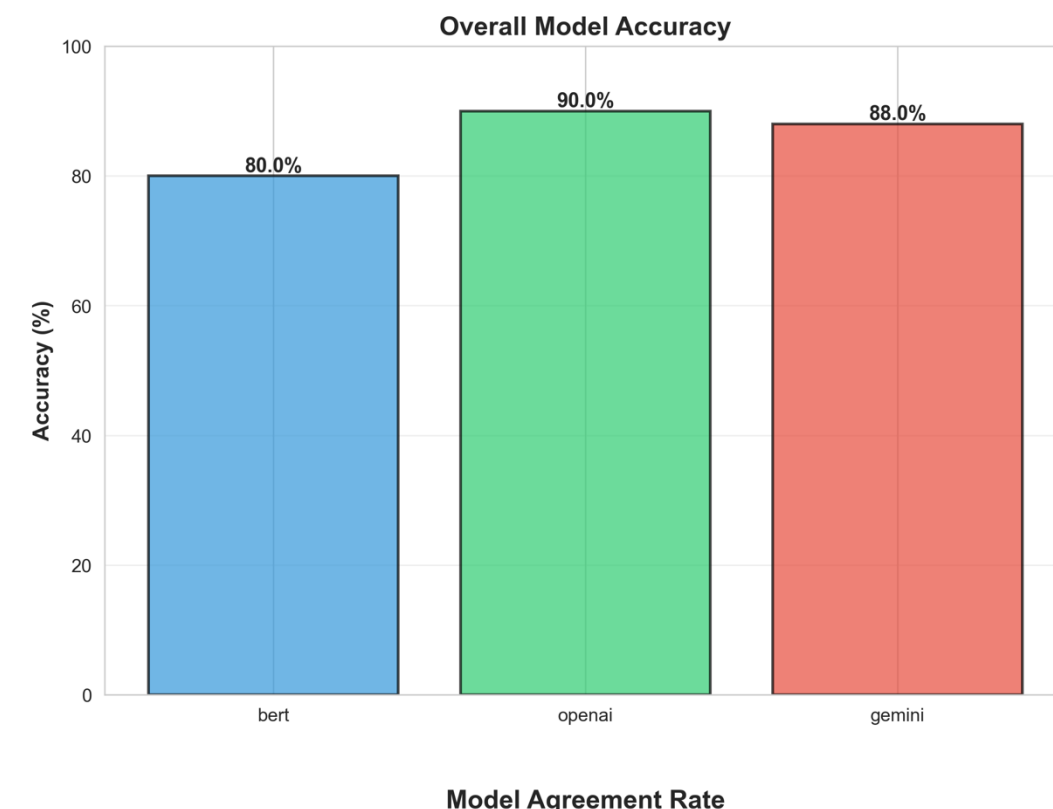
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ALGORITHMS

R1 RESULTS — EMOTION CLASSIFICATION

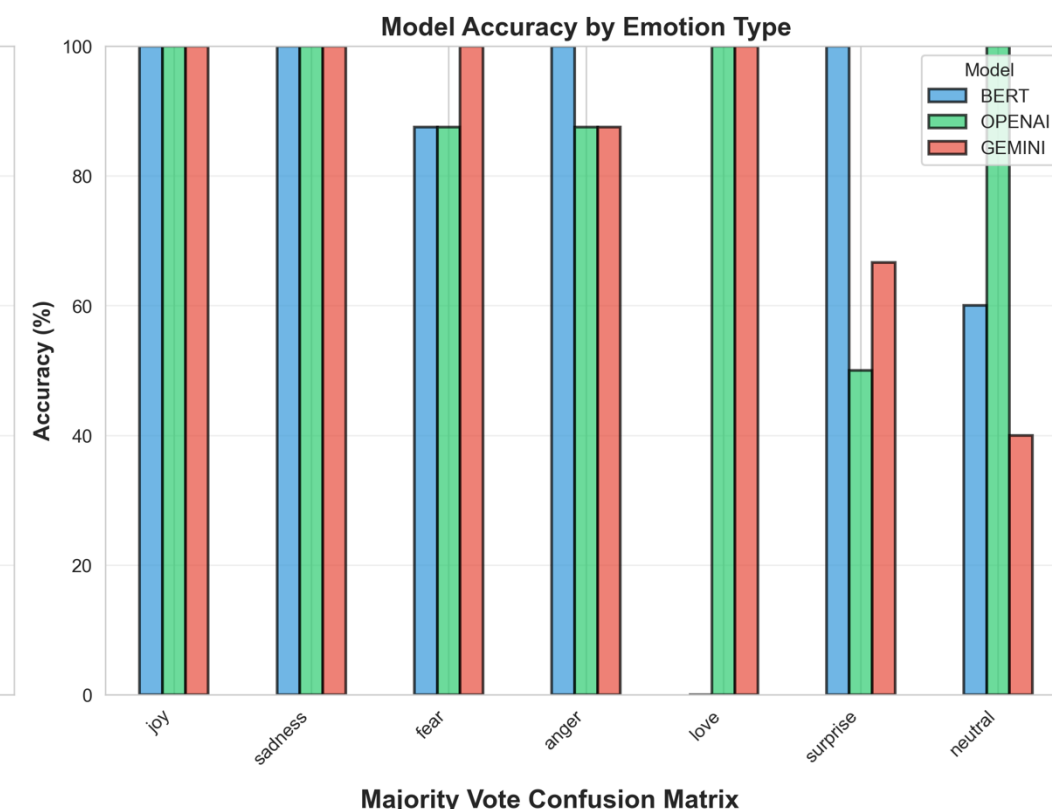
80%

BERT ACCURACY

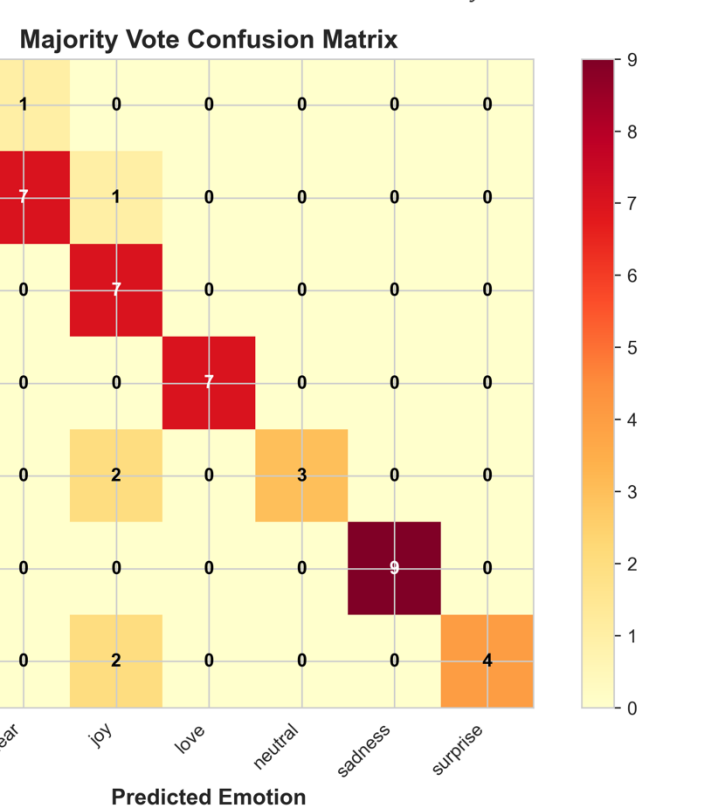
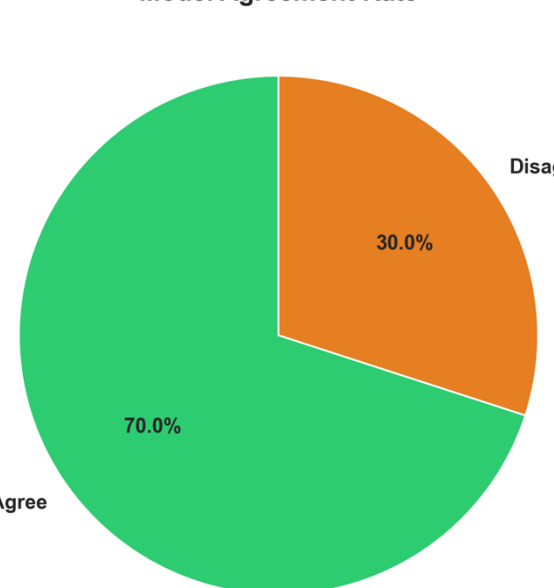


90%

GPT-4o



Model Agreement Rate



✓ H1 Confirmed ($\geq 75\%$)

- 5 of 7 emotions: 87–100%
- Cross-validated with GPT-4o and Gemini
- Love harder due to overlapping vocabulary

R2 RESULTS — RECOMMENDATION

R2 Algorithm Comparison (Including Random Baseline)

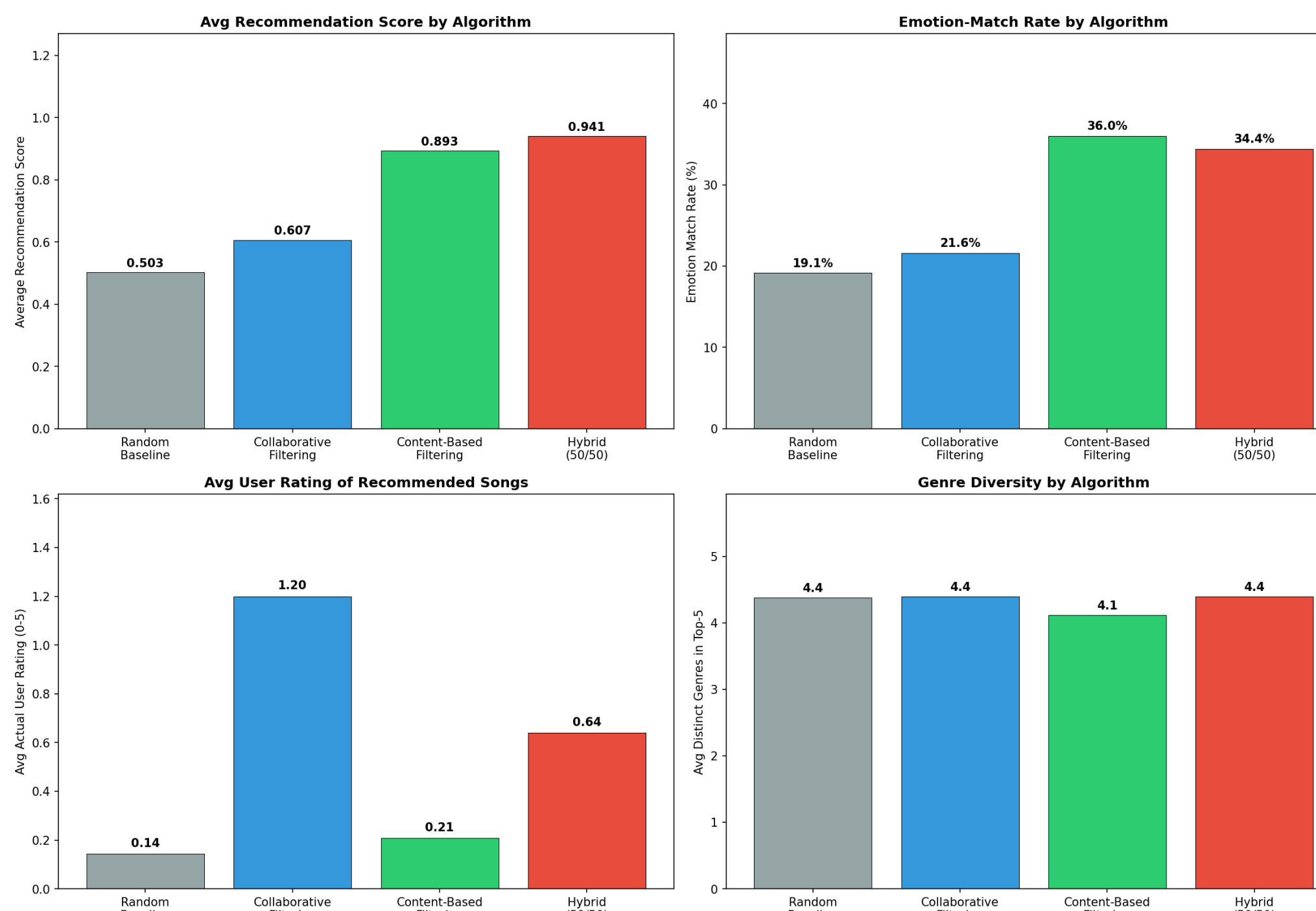
Algorithm	Score	p-value
Random	0.503	—
Collaborative	0.607	< 0.001
Content-Based	0.893	< 0.001
Hybrid (50/50)	0.941	< 0.001

✓ H2 Confirmed — Hybrid outperforms all

Content-based → best emotion alignment

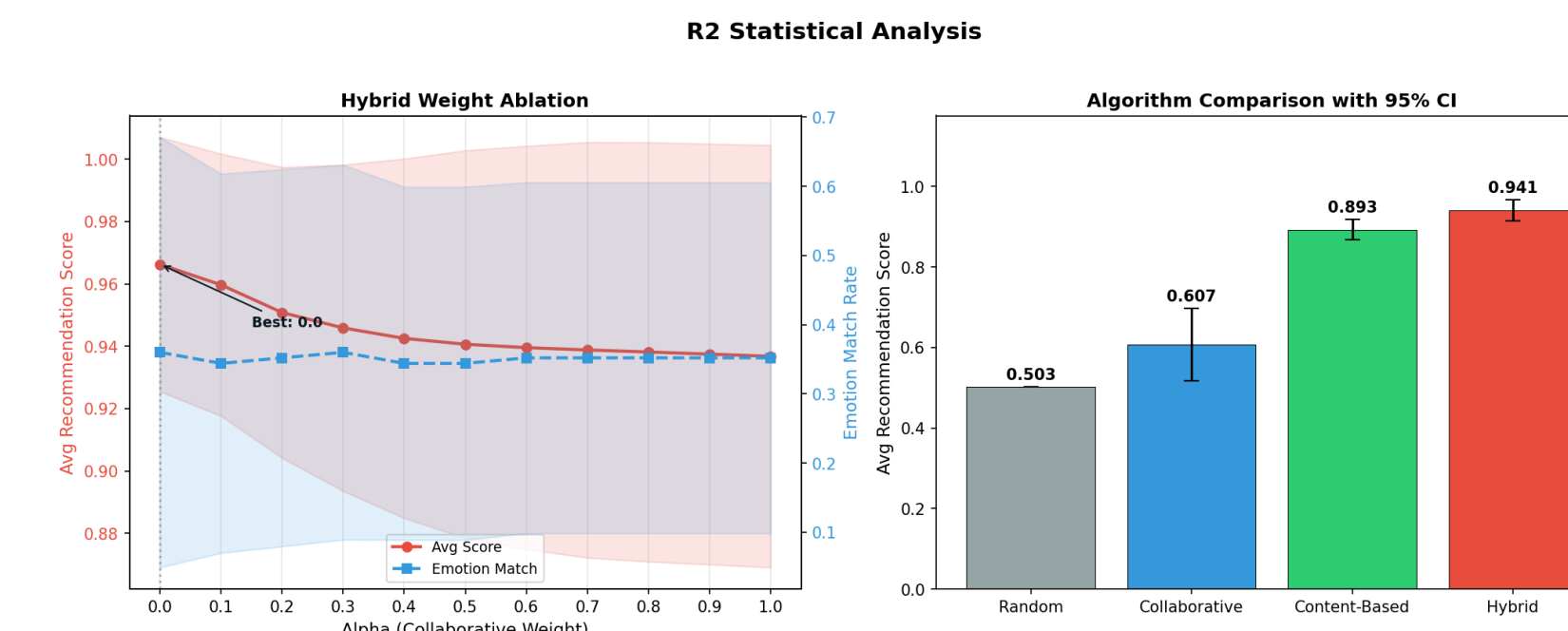
Collaborative → best user ratings

Hybrid captures both strengths



STATISTICAL ANALYSIS

All comparisons: $p < 0.001$



NOVEL CONTRIBUTIONS

1. Transformer NLP for Autobiographical Memory

Applied transformer-based emotion classification to personal narratives written in older adults' voice — a domain not previously explored with this approach.

2. Systematic Emotion-to-Music Feature Mapping

Created a principled mapping from seven emotion categories to musical features (tempo, valence, energy, mode) grounded in music psychology research.

3. Emotion-Preserving Hybrid Recommendation

Optimized hybrid recommendation to preserve emotional alignment rather than maximizing popularity — achieving 0.941 average score with statistical significance.

DISCUSSION & IMPLICATIONS

- Love is systematically misclassified as joy/sadness by BERT; GPT-4o and Gemini correctly classify all love stories
- High-confidence errors (5/10 ≥ 0.80) indicate systematic model bias, not random noise
- GPT-4o (90%) and Gemini (88%) outperform BERT (80%), suggesting larger context windows help disambiguate
- Neutral stories with positive undertones trigger false joy predictions (60% accuracy)
- Content-based: highest emotion-match; collaborative: highest user rating — complementary strengths justify hybrid

CONCLUSIONS

- ✓ H1 CONFIRMED: DistilRoBERTa achieves 80% accuracy, exceeding the 75% threshold
- ✓ H2 CONFIRMED: Hybrid (0.941) outperforms collaborative (0.607) and content-based (0.893). All $p < 0.001$
- ✓ Transformer NLP can reliably extract emotions from autobiographical narrative text
- ✓ MelodyMemo establishes a computational foundation for emotion-preserving music recommendation

FUTURE WORK

- Pilot validation with older adults in music therapy settings
- Continuous emotion modeling for complex emotions
- Adaptive personalization using user feedback

KEY REFERENCES

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